ENEG11009

The Weekly Notes

Authored by
Lachlan J. W. Wyer

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Week 1 - Energy & Energy Conservation

Energy:

Energy is the ability to work, either physical or mental. It does not matter if this work is self-contained or used up in a system. In a closed system, there is a fixed about of total energy within the closed system, that does not interact with anything outside of the system. The main SI unit for energy is Joules (J).

Types of Energy:

Kinetic Energy (E_{KE} or K_E):

This energy is the energy of motion. Implying that all moving objects contain a form of K_E, e.g. The wind or water.
As shown in K_E's equation, the heavier or faster an object is, the greater the its K_E. As such, kinetic energy comes in two main forms:

- Translational K_E: The motion along a specified path
- Rotational K_E: The motion around an axis

As such the formula for kinetic energy can be derived from both the object mass and velocity.

$$K_E \; \alpha \; m \;\; \& \;\; K_E \; \alpha \; v^2$$

$$\therefore K_E = \frac{1}{2}mv^2$$

- K_E - Joules
- m - Mass of the object(kg)
- v - velocity of the object ($\frac{m}{s}$)

Potential Energy (E_{PE}):

This is a type of stored energy that can readily be converted into another form. For example the water store behind a dam, or thermal vents. There are three main types of potential energy: gravitional, elastic and thermal.

Gravitional Potential Energy (G_{PE}):

This type of energy is the type of stored energy associated with objects height & mass.

$$E_{PE} = mgh$$

- G_{PE} - Joules (J)
- m - mass of the object, can also be calculated from an object's forces ($F = mg$)
- g - Standarized gravity ($9.8m\ s^{-1}$)
- h - Height (m)

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Elastic Potential Energy (E_{PE}):

$$E_{PE} = \frac{1}{2}kx^2$$

- E_{PE} - Joules of energy
- k - The spring of elastic constant ($\frac{N}{m}$)
  - This constant can also be calculated($k = -\frac{F_s}{x}$)
- x - The displacement of the object(m)

Thermal Energy (E_{TH}):

This energy is created due to movement of particles. The hotter a substance, the more its particles move. Also, the greater its friction co-efficient the more the substance will heat up, and thus the E_{TH} will increase.

$$E_{TH} = f_F x$$

- E_{TH} - Measured in Joules (J)
- f_F - The friction created in movement ($f_F = \mu mg$)
- x - The distance moved an object

OR

$$E_{TH} = \mu mgx$$

- μ - The friction coefficient of the floor
- m - The mass of object (kg)
- g - Gravitional Constant ($9.8m\ s^{-1}$)

Force:

Force can be split into two main types:

1. Conservative
    - Independent of a Path
    - Contains and stores useful energy
2. Non-conservative
    - Dependent on a specific path
    - Cannot store useful energy

Energy Transformations:

According to Newton's First Law, energy can neither can be created nor destroyed, only transformed from one form to another. As such, the total energy remains the same within a closed system. In the end it will only be converted into undesirable forms. Energy can be transferred between a system and its environment, through two main ways.

$$E_{Total} = K_E + E_{PE} + G_{PE} + E_{TH} + \ldots + W$$

1. Work: Mechanical transfer of Energy, e.g. Pushing or Pulling
2. Heat: Non-mechanical transfer of Energy, e.g. A temperature difference between objects

Work:

Work is done when a force is applied over a distance.

$$W = \vec{F} \cdot \vec{d}$$

- W Work is a scalar product (J or $N \cdot m$)

- The vectors $\vec{F}$(N) and $\vec{d}$(m) must ve in the same direction or have them resolved into a parrallel directions.

E.g. A crate is 500cm with a force of 500N

$$\begin{aligned} W &= \vec{F} \cdot \vec{d} \\ &= 500 \cdot 500 \times 10^{-2} \\ &= 2500J \\ &= 2.5kJ \end{aligned}$$

Power:

Power can be described as an amount of energy convertted (the work done) per unit of time. The unit of time is usually seconds.

$$P = \frac{W}{t} \text{ OR } P = \frac{E}{t}$$

The SI unit of P is $\frac{J}{s}$ or W(Watts).
A different way of calculating an instaneous output:

$$P = \frac{W}{t} = \frac{\vec{F} \cdot \vec{d}}{t} = \vec{F} \cdot \frac{\vec{d}}{t} = \vec{F} \cdot v$$

Efficiency:

Inessence, basically nothing can possibly be 100% efficient. Reductions can be found in:

- Process limitations: Practical implementation within the realworld
- Fundamental limitations: The laws of physics and the realworld

A system can have lower efficienies can be due to increase in energy losses.

$$\eta = \frac{UsefulEnergyTransferred}{TotalEnergySupplied} \times 100$$

- η - The efficiency of a system (%)

Week 2 - Heat & Gas

0th Law of Thermodynamics:

This law states that two systems can be considered in thermal equilibrium, iff there is on net flow of E_{TH} between the systems when a path is connected that allows flow. When two systems in thermal equilibrium can be assumed to have the same temperature.
E_{TH}: The internal energy of a system due to temperature.
Temperature: Average K_E of the atoms that make up an ideal gas.

Temperature Scales:

$$^oC -> ^oF : \frac{9^o}{5} C + 32$$

$$^oC -> K : {}^oC + 273$$

The Kelvin and Celsius scales are considered to be almost thermally equal. Meaing that when one is change by an amount (e.g. 1K) the other change by the same (e.g. 1^oC).

Heat (Q):

Heat is considered to be the energy transferred across the boundary of 2 different systems because of a temperature difference. Heat has different methods of exchanging energy with objects & their surrounds:

- Conduction: Mostly in solids without the movements of mass
- Convection: Generally in liquids or gases, and the transfer involves movement of mas
- Radiation: Does not need a medium to transfer heat, e.g. heat can be transferred in a vaccum
- Evaporation; Molecules movings through states, e.g. Liquid -> Gas below the molecules boiling point

Conduction

The transfer of thermal energy between connected materials.

$$Q \; \alpha \; \Delta T, \text{where } \Delta T = T_H - T_C$$

$$Q \ \alpha \ A, \text{where } A = CSA$$

$$Q \ \alpha \ \frac{1}{L}, \text{where L is the length between molecules}$$

As such the equation for conduction is as follows:

$$P_{cond} = \frac{Q}{t} = \frac{kA\Delta T}{L}$$

- P_{cond} - In Watts
- k - The thermal conductivity of the object ($\frac{W}{m} \cdot K$)
- A - Is the the CSA
- ΔT - The temperature difference
- L - THe Thickness between the molecules or length of the object

Radiation

$$P_{rad} = \frac{Q}{\Delta T} = e\sigma A(T_1^4 - T_2^4)$$

- e - The emmisivity of a particular surface (0->1)
- σ - The Stefen-Boltzmann constant ($5.67 \times 10^{-3} \frac{W}{m^2} \cdot K^4$)
- A - CSA
- T_1^4 or T_2^4 - The temperature measured in Kelvin

Specific Heat Capacity:

The quantity of heat absorbed to increase 1kg of substance increase 1K.

$$Q = mc\Delta T$$

Atomic Model of Matter:

Solid:

- Particles are connected with strong, spring-like bonds
- The particles have a defined shape
- Can be compressed slightly

Liquid:

- Weak bonds present between particles
- Particles can move slightly
- Takes the shape of the container that it is placed in, e.g. bowl.

Gas:

- No inter-molecular bonds are present
- Particles are in random motion
- Particles interact with elastic collisions
- Move freely & fill the entire vessel they are placed in

When molecules change phase, heat is still being transferred with no temperature change. The amount of heat required for this called latent heat.

Latent Heat:

There are two types of latent heat, 1. Heat of Fusion and 2. Heat of vaporization. Heat of fusion (L_F) is the amount of heat required to convert all 1kg of substance from a solid to a liquid. Heat of vaporization (L_v) is similar except it requires changing from a liquid to a gas. The SI unit for this is $\frac{kJ}{kg}$.

$$Q = mL$$

- Q - Heat required to change phase. This can either be $\pm$
- m - The mass of the substance changing states
- L - The Latent required to change state

1st Law of Thermodynamics:

The change in energy is the difference between the heat added & the work done.

$$\Delta U = U_f - U_i = Q - W$$

If $Q > 0$, heat has been added to system. Else $W > 0$, work has been done by the system.

2nd Law of Thermodynamics:

Entropy (s) always increases, as it is an irreversible process. Everything in order will always lead to disorder.

$$\Delta s = \frac{\Delta Q}{T}$$

Energy Reservior:

A system with a huge heat capacity (whether hot or cold) that does not change in temperature when a change in heat happens. For example, adding 1L of boiling water to the ocean does not "really" change the temperature of the ocean. Even though alot of heat has been added into the ocean.

Pressure:

The value of pressure is a force applied $\perp$ to the surface area of a contianer or vessel. The SI unit is $1Pa = 1\frac{N}{m^2} = 1\frac{kg}{m\ s^{-2}}$. Other unit conversions to remember are $1atm = 101,300Pa = 14.7psi$.

$$p\ \alpha\ T, \quad p\ \alpha\ \frac{1}{V}, \quad p\ \alpha\ n$$

$$pV = nRT$$

- p - Pa
- V - m^3
- n - moles of substance
- R - The real gas constant ($8.31\ \frac{J}{mol} \cdot K$)
- T - Temperature in Kelvin

Week 3 - Electricity

SI Units used:

Quantity	Name (Base Unit)	Symbol
Length	metre	m
mass	kilogramme	kg
Time	second	s
Electric Current	**Ampere**	**A**
Thermodynamic temperature	Kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Amps:

The charge of an electron is $1.602\ 176\ 634 \times 10^{-19}$C. The amps in a circuit can be defined as the quantity of charge passing through a cross-section per unit time.

$$I = \frac{Q}{t}$$

- Q - The unit for measuring charge is Coulomb (C)
- t - Time (seconds)

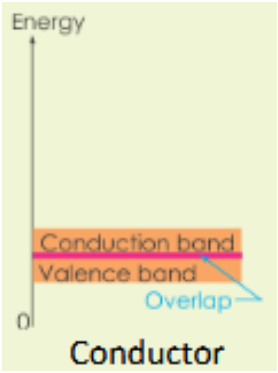
Voltage:

The voltage of a system is the difference in potential between elements in the circuit. Voltage is related to Amps through it having $\frac{\text{Joules}}{\text{Coulomb}}$.

$$V = \frac{W}{I} = \frac{J/t}{I} = \frac{J}{It}$$

Conductors:

Conductors are a material that allows free movement of charge. For example allows current to flow through it, like every metal is basically a conductor. Conductors work through having the conduction and valence electron bands overlapping. This means that little to no energy is required to change bands.



Insulator:

Insulators are materials that prevent the free movement of charge. Examples of insulators are rubber, ceramic, cork, and air. They work through the conduction and valence bands having an energy gap. This is because to change energy levels require large amount energy.

Semi-conductors:

Semi-conductors are a crystalline material that has properties between a conductor and insulator.

Resistance:

The resistivity of a material is mesaured in how strongly it resists an electrical current.

$$\sigma = \frac{1}{\rho}$$

The resistance of a material is a property of a specific wire that depends on of factors.

$$R \, \alpha \, L, \quad R \, \alpha \, \frac{1}{A}$$

$$R = \frac{\rho L}{A}, \quad \rho = \frac{RA}{L}$$

- R - Ohms (Ω)
- ρ - Restivity ($\Omega \cdot m$)
- L - Length of material (m)

- A - CSA of material (m^2)

Wire size AWG	Current capacity copper wire (amps)	Resistance per 1000 feet (ohms)	Resistance per km (ohms)
16	15	4.016	13.176
14	20	2.525	8.284
12	25	1.588	5.210
10	30	0.9989	3.277
8	40	0.6282	2.061
6	55	0.3951	1.296
4	70	0.2485	0.815
3	85	0.187	0.614
2	95	0.1563	0.513
1	110	0.1239	0.406
1/0	125	0.0983	0.323
2/0	145	0.0779	0.256
3/0	165	0.0618	0.203
4/0	195	0.049	0.161

Electrical Circuit Elements:

The voltage values within a circuit should follow the positive sign convention. $+ \rightarrow -$ voltage, there is considered to be a voltage drop between the two points. For current, the current coming into the node has to equal the current leaving.

Ohm's Law:

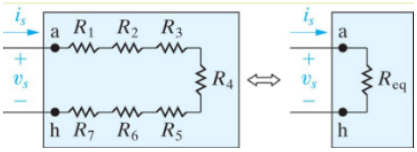
According to this law, the current flowing through a material is proportional to its voltage, e.g. $I \ \alpha \ V$.

$$\therefore I = \frac{V}{R} \ \ OR \ \ V = IR \ \ OR \ \ R = \frac{V}{I}$$

Equivalent Resistance:

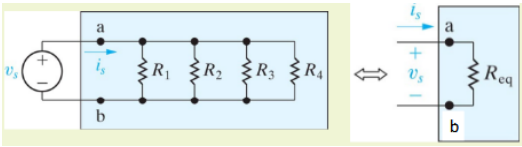
Things to remember:
If a circuit is connected in series, all elements have the same current and share the voltage.

$$R_{eq} = \sum_{i=1}^{n} R_i$$



Whereas, connected in parrellel, they contain the same voltage and share current.

$$Req = \frac{1}{\sum_{i=1}^{n} \frac{1}{R_i}} = (\sum_{i=1}^{n} \frac{1}{R_i})^{-1}$$



Kirchoff's Laws:

Current (KCL)

The sum of all within a node must equal 0. Therefore, the current entering a node must be equal to that exiting.

$$\begin{aligned} -I_{In} + I_{Out} &= 0 \\ \sum I_{In} &= \sum I_{Out} \end{aligned}$$

Voltage (KVL)

The sum of all voltage differences within a circuit (or a specific closed loop) must be 0.

$$\sum_{i=0}^{n} V_i = 0$$

Week 4 - Rotational Motion & AC Power

Rotational Motion:

$$\theta(\text{rad}) = \frac{s}{r}$$

This is the defition of a radian. Remember Specialist and Methods converstions between radians and degrees. It is important to remember that rotational motion

happens about the turn of a central/interal axis. Rotional motions comes in two forms:

1. Linear Motion
   $x = v_0 t + \frac{1}{2} a t^2$, where:
   - x - The displacement along an axis
   - v_0 - An object's initial velocity
   - a - The acceleration of an object
   - t - Time (secs)
2. Angular Velocity
   $\omega = \frac{\Delta\theta}{\Delta t}$, where:
   - ω - $\frac{rad}{s}$
   - $\Delta\theta$ - Change in radians
   - Δt - Amount of times to take the change

The angular velocity is just the derivative with respect to time of the angular position. Other units can also be used, e.g. RPM, but must be converted to $\frac{rad}{s}$ for these equations. For rotation motion, the time to complete one rotation is the period of motion (T). Compared to the number of revolution per seconds is the frequency (f).

$$f = \frac{1}{T}$$

$$\omega = 2\pi f$$

Angular Speed:

$$v = r\omega$$

Angular Displacement:

Linear Displacement

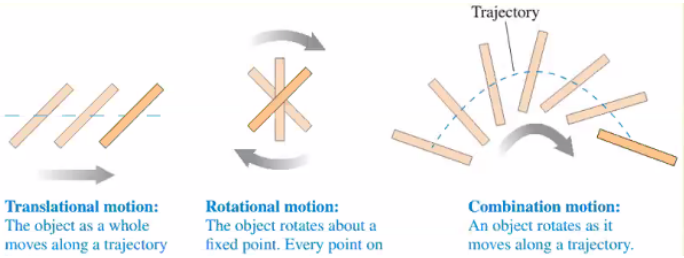
$$x_f - x_i = \Delta x = v_x \Delta t$$

Angular Displacement

$$\Delta\theta = \theta_f - \theta_i = \omega \Delta t$$

Rotation of a Rigid Body:

A rigid body is an object that does not change shage when motion is being taken. For example, types of rigid body motion is shown in the image below.



Just some equations to understand

SYNTHESIS 7.1 Linear and circular motion

The variables and equations for linear motion have analogs for circular motion.

	Linear motion	Circular motion
Variables	Position (m) x	Angle (rad) θ
	Velocity (m/s) $v_x = \frac{\Delta x}{\Delta t}$	Angular velocity (rad/s) $\omega = \frac{\Delta\theta}{\Delta t}$
	Acceleration (m/s^2) $a_x = \frac{\Delta v_x}{\Delta t}$	Angular acceleration (rad/s^2) $\alpha = \frac{\Delta\omega}{\Delta t}$
Equations	Constant velocity $\Delta x = v\Delta t$	$\Delta\theta = \omega\Delta t$ Constant angular velocity
	Constant acceleration $\Delta v = a\Delta t$	$\Delta\omega = \alpha\Delta t$ Constant angular acceleration
	$\Delta x = v\Delta t + \frac{1}{2}a(\Delta t)^2$	$\Delta\theta = \omega_i\Delta t + \frac{1}{2}\alpha(\Delta t)^2$

Tangential (Linear) Acceleration:

$$a_t = \frac{\Delta v}{\Delta t} = \frac{\Delta(\omega r)}{\Delta t} = \alpha r$$

Centripital acceleration is $\perp$ to its $\vec{v}$ & aims towards the centre.

Torque:

Torque is the measure of the effectiveness of a force at causing an object to rotate around a pivot.

$$\tau = r \cdot F sin(\phi) = -r_\perp mg$$

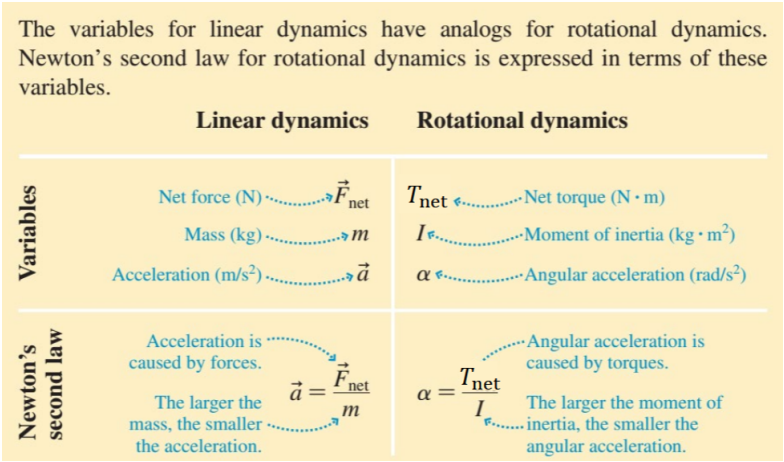
- τ - Torque $(N \cdot m)$

- r - Distance (m)
- F - Force (N)
- ϕ - Angle of movement, required as the force must be $\perp$ to the object or resolved

The $\tau_{net} = \sum T = \alpha I$, where I is the moment of ineritia.

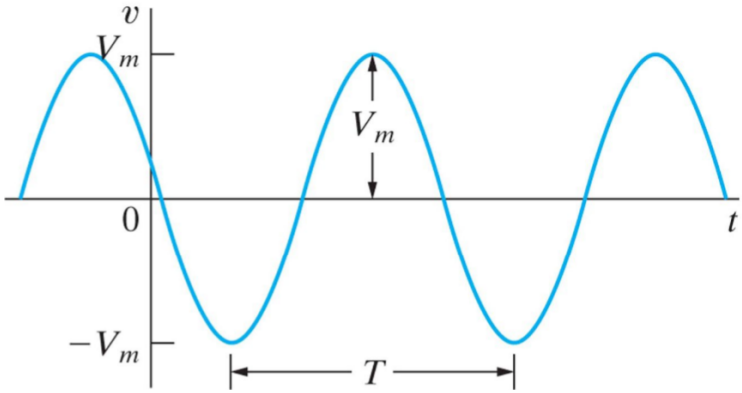
$$I(kg \cdot m^2) = \sum m_i r_i^2$$

And Newton's 2nd Law of rotational motion states that $\alpha = \frac{\tau_{net}}{I}$.
As such, some more rotational dynamics items to know are:



AC Power:

Power can be converted between AC and DC power profiles. DC is a constant voltage overtime, e.g. 12V DC is always 12V. Whereas, 12V rotates between +12V & -12V, because AC alternates polarity overtime.
The default AC waveform is the sinusoidal waveform.

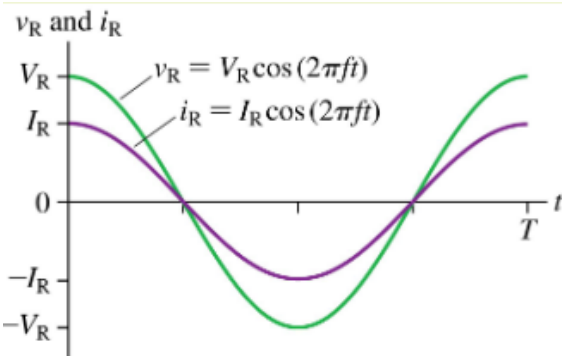


$$v(t) = V_m sin(\omega t + \psi) = V_m cos(\omega t + \phi)$$

The phase angle (ψ, ϕ) defines the value of v(t) @ t=0. Positive values shift the graphs left, whilst negative values shift right.

Resistors:

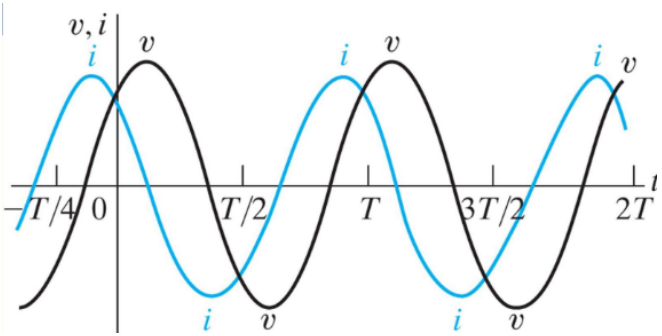
In a resistor circuit: $v_{source}(t) = v_R(t)$.
Here max amps and max voltage happens at the same time. With this it could be said that current is in phase with the voltage.



Capacitor:

$$v_C(t) = V_C sin(2\pi f t), \quad \text{Assuming that } \phi = 0$$

In the capacitor graph, the current in leading the voltage by $\frac{\pi}{2}$.



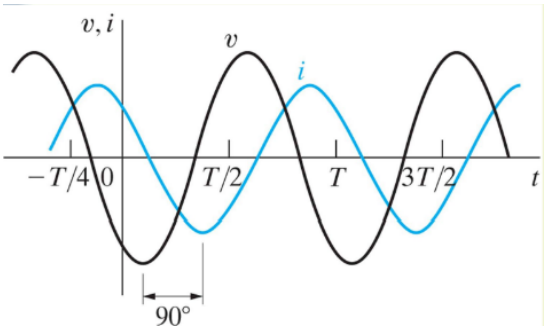
$$X_C = \frac{1}{\omega_C} = \frac{1}{2\pi f tc}$$

Inductor:

For this, assume that the inductor is connected too the wave source.

$$i_L = I_L cos(2\pi f t)$$

Here the current is lagging the voltage by $\frac{\pi}{2}$



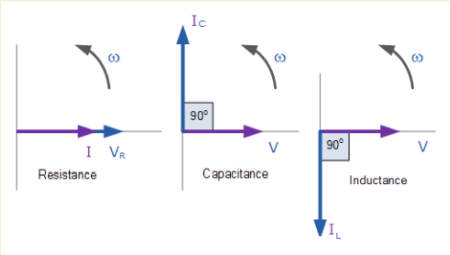
$$X_L = \omega L = 2\pi f L$$

Relationships:

To remember the units and styles see the image below.

R: Ohm
C: Farad
L: Henry

R: Ohm
X_C: Ohm
X_L: Ohm



Root Mean Square (RMS):

The RMS value is useful because AC can then be compared DC directly. Thus, AC RMS values can be compared against DC values, creating a heating a value (e.g. power and heat).

Resistor:

$$V_{RMS} = I_{RMS}R$$
$$v_{RMS}(t) = I_{RMS}(t) \cdot R$$

Capacitor:

$$V_{RMS} = I_{RMS} \times \frac{1}{Q_C}$$
$$v_C(t) = I_C(t) \cdot X_C$$

Inductor:

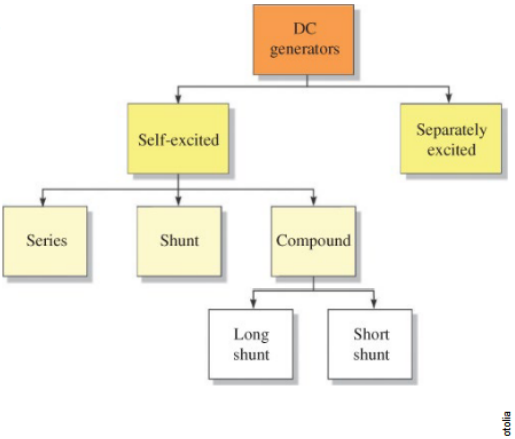
$$V_{RMS} = I_{RMS}\omega L$$
$$v_L(t) = I_L(t) \cdot X_L$$

Week 5 - Generators:

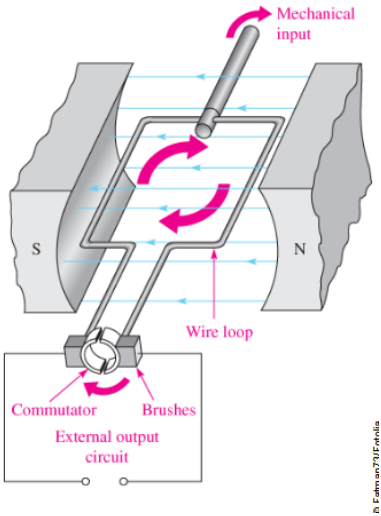
DC Generators

All DC generators have a voltage induced across a conductor, where an inductor through a magnetic field. This phenomeoa is calld electromagnetic induction. To cause this relative motion is all that is needed. Whether the

conductor or the magnet can move. The official name for this is called Faraday's Law.
There are several ways in which DC generators can be classified. For example, the simpliest DC generator, powered through a permenant magnet.



Some points to consider are that wil the self-exicted generator, the magnetic field is created through power the generated has made. Compared with a seperately excited generator. Where external power has been supplied.
A note with this is that a shunt generator can be thought of as being in parrellel with itself.
The basics of a generator is shown in the image below. Here the wire loops are wrapped around a shaft (usually iron), which is rotated in the magnet. Power is connected through the brushes only when they are in contact with the communtator. In this example when the wire rotates clockwise, voltage is induced when conductors cut the magnetic flux field. ∴ the maximum power is generated when the conductors are ⊥ & a minimum amount when ∥.

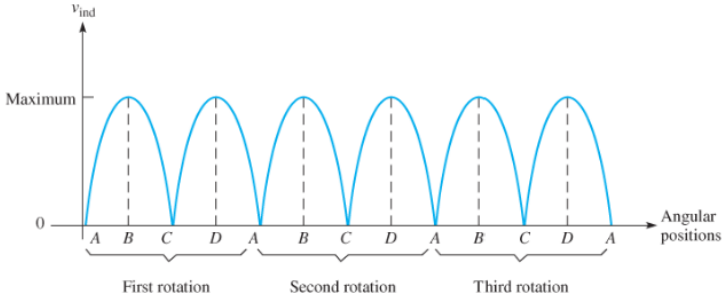


To calculate the induced:

$$V_{ind} = B_{\perp} l v$$

- V_{ind} - Induced Voltage
- $B_{\perp}$ - Flux field density ($\frac{\phi}{A}$ in $\frac{Wb}{m^2}$ or Telsa(T)
- l - length of the wire in the windings (m)
- v -Velocity of the motor ($m\ s^{-1}$)

Graphing this versus its angular position, creates the humps below. To stabilize this into the sterotypical DC voltage, flat line, more wire windings can be used.



The armature is where the physical voltage is induced. The communtator is where the windings are connected, and copper is used to isolate each winding. If a brush is use, frequent maintainence is required. E.g. to clean the communtator or replace the brush. This is because the brush is made up of carbon.

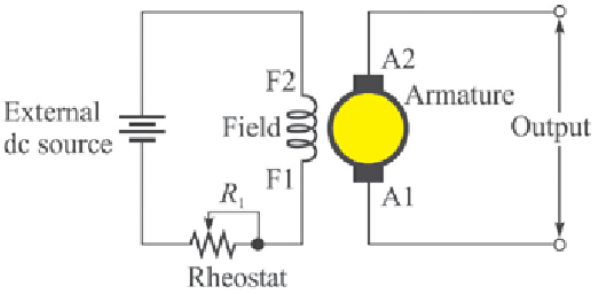
Stator Assembly:

Field coils (field elctromagnets) generate the magnetic field in the generator. How the coils recieve current determine their classification. If the generator creates its own current, it is self-exicted. This current is called the excitation current.

DC Generator Types:

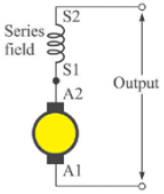
Sperately Excited

The current is provided externally. The more amps the wire recieves the greater is magnetic flux. The rheostat is a variable resistor controls the charge current.

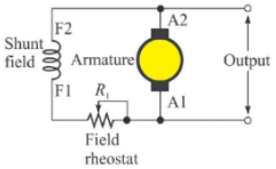


Self-excited

The difference between a series and shunt generator is how the armature & field winding are connected. How it is able to generate enough current to start workings is through a bit of residual magnetisium. Once enough voltage is created to power the windings, power is then generated.



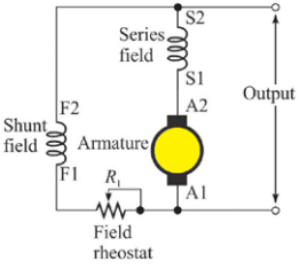
- Series



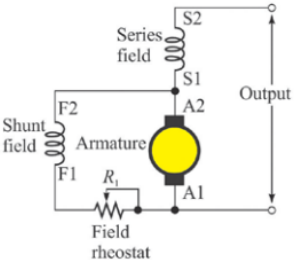
- Shunt (Parallel)

Compound Generator

In compound generators there is two field windings (one series & one shunt). Depending on where the windings one placed is whether it is a long or short shunt.



- Long shunt



- Short shunt

Polarity of DC Generators:

The are two main ways to change voltage in a DC generator:

1. Changing rotation direction of the prime mover. This changes the armatures voltage.
2. Changing the field winding with respect too the armature winding.

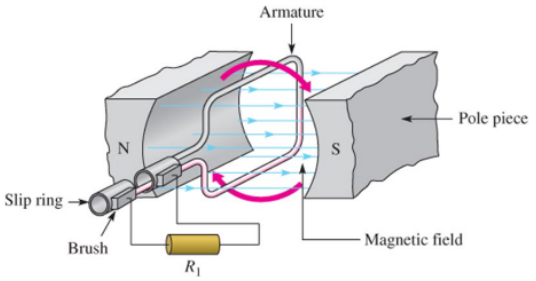
AC Synchronous Generators:

A synchronous generator is called so because its generated voltage waveform is in sync with the rotation of the generator. There are two main types:

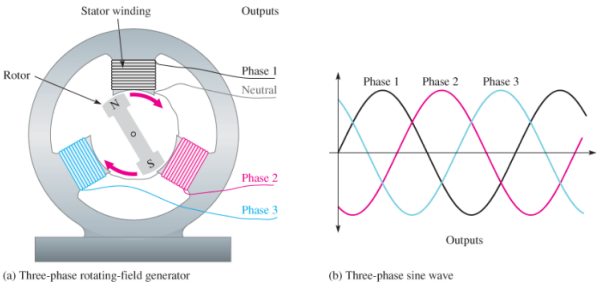
1. Rotating Armature
2. Rotating Field

Rotating Field Type

Magnetic field is stationary & supplied by elctromagnets. The output power is taken from the rotor through the brushes and slip rings. All large synchronous generators are rotating field types.



3-Phase Generator



The frequency of a synchronous generator is determined by the number of poles and speed of rotation.

$$f = \frac{N_p \times RPM}{120}$$

- f - Frequency (Hz)
- N_p - The number of poles present in the generator
- RPM - Revolutions per minute that is does

4-Pole Generator

In this generator, 1 revolution produces 2 cycles of the induced voltage waveform.

AC Induction Generators:

The AC induction generator is also known as an asynchronous generator. This is because as the waveform in not in sync to the rotational speed.
A bit to understand with this is that this generator can function as either a motor or generator. This is that as the motor's rotor spins at a lesser speed than the synchronous speed, whereas an asynchronous generator's rotor spins faster than the synchronous speed.

Weeks 6/7 - Solar Energy:

Photovoltaics:

These systems are made with silicon semi-conductors.

Semi-conductors

Semi-conductors are made out of crystalline silicon (c-Si). Silicon is an element contains a valence band that can possibly have 8 electrons. With this, they will generate a charge with these electrons.

N-Type Silicon

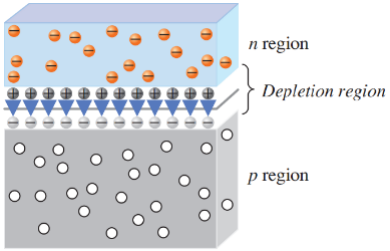
When doping silicon with a Group V element. 4 out of 5 valence electrons to form covalent bonds. This makes the semi-conductor slightly negatively charge. Usual doping amount is 100 million Si to 1 Group V element.

P-Type Silicon

Here silicon is doped with a Group III element. Creating a slightly positive charge. The missing electron creates what is called a hole.

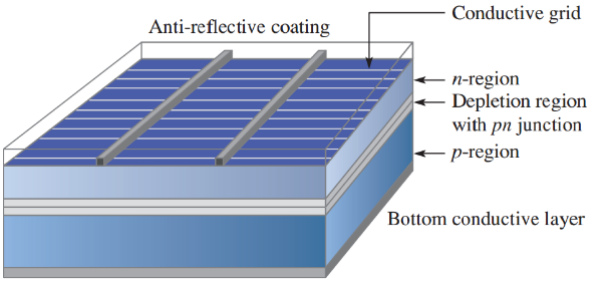
PN Junctions

This junction is formed through a single silicon atom layer, with n-type & p-type on either side. An example of this is shown in the image below. It is around the depletion region charges can freely move around. This junction is crucial in diodes as the current can only pass in one direction.



PV Cell:

An example basic PV cell structure is shown below. When light enters the cell, if the photon energy is greater than the energy gap (1.12eV), a charge will be created.



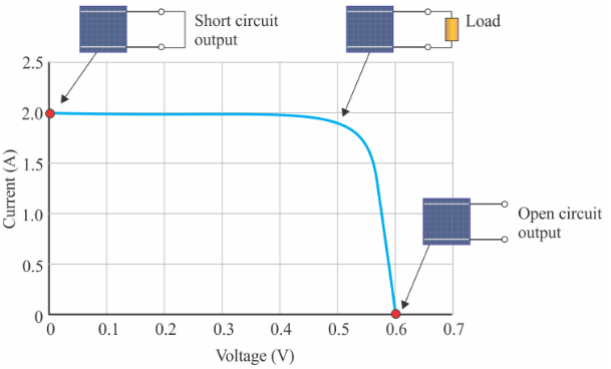
Types of PV Cells:

There are three main types of PV cells:

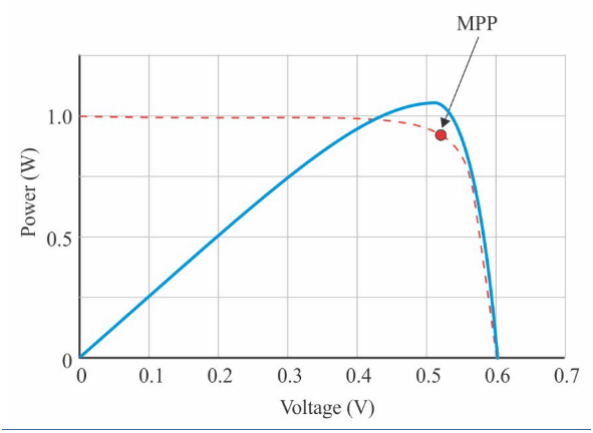
1. Monocrystalline - Single cell waffers, providing the highest efficiency
2. Polycrystalline - Composed of smaller crystals, slightly less efficient
3. Thin-film - Made of amorphous, the lowest cost to manufacture and least efficient

Characteristics of PV Cells:

The I-V characteristics is constant a variety of supplied voltages for a specified incident light energy.



The max power output occurs on the "knee" of the curve.



On the back of each module is a junction box, containnig two standardized cables. These connect the modules. When connecting modules in series it is called a string. Multiple different strings can be connected in parrallel to form an array.The whole system is a combiner.

Solar Power System:

Stand-alone systems do not power the grid. Rather are used for smaller, remote applications. Most systems are only DC @ 12, 24, 48V. When using a solar cell to charge a battery, a charge controller is used to regulate & limit the charging rate. So as to not over charge the battery.

Sizing the System:

$$P_{array} = \frac{W}{t_{solar}\ \eta_{sys}}$$

- P_{array} - W
- W - Wh
- t_{solar} - Number of hours of solar a day
- η_{sys} - Efficiency of the system

$$Ah = \frac{W_{day}\ t_{store}}{V\ B_{DoD}\ \eta_{inv}}$$

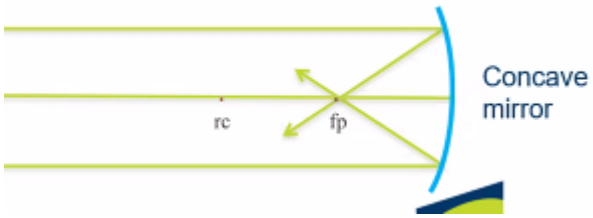
- Ah - Amphours for the battery
- W_{day} - $\frac{Wh}{day}$
- t_{store} - The number of days the battery needs to provide power (days)
- V - Dc system voltage to the inverter
- B_{DoD} - Batteries max depth of discharge
- η_{inv} - Efficiency of the inverter

$$FF = \frac{P_{MPP}}{V_{OC}\ I_{SC}}$$

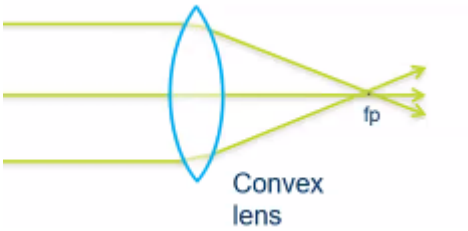
This describes how close the PV cell is to the theoretical maximum efficiency.

Solar Concentrators:

A SC uses either mirrors or lenses to focus the light upon a point. A mirror system assumes the light comes in parrallel to the mirror. This is then reflected to a point on the axis called the focal point(f_p). The f_p is always $\frac{1}{2}r_c$.



When the light passes through a lense, a convex one is used. Because of costs, lense systems are not widely used.



Concentrated Solar Power:

CSP system can be used to store thermal energy in a fluid, then use it whenever it is needed. It works similar to the traditional steam engine.

Solar Hot Water Systems:

Solar hot water systems work through fluid-plate collectors, focusing sunlight on a manifold to heat the water. The manifold's pipes are usually coverede in a material that has a high absorbance. In areas subject to freezing a closed-loop, pressurized glycol-water system will have to be used.

Solar Movement:

Remember the earth GPS with latitudes and longitudes. The earth is on a 23.4^o include to equator. It is important to know this to understand the amount of sunlight on the earth.

Solar Tracking

Solar cells work best with direct sunlight. Trackers follow the Sun's direction throughout the day. Increasing the time that the cells can achieve maximum efficiency. Single axis tracking only does East-West movement. Whereas dual axis can also move North-South. Within these there are two types of trackers.

1. Active - Using actual motors
2. Passive - Uses liquid & gas properties to move the cell

Other Equations to Know:

$$\tau = K\phi I_a$$

$$\tau = \frac{P}{\omega}$$

- τ - Torque ($N \cdot m$)
- K - $N \cdot m$ or $Wb \cdot A$
- ϕ - Wb
- I_a - A in the armature
- P - Power (W)
- ω - $\frac{rad}{s}$

$$V_S = C_{emf} + I_a R_a$$

- V_S - The applied voltage
- C_{emf} - Counter EMF (V)
- $I_a R_a$ - Their respective items in the armature

Week 8 - Wind Energy:

Air Pressure:

The pressure of the air (air pressure) is crucial to the wind systems.

$$P = \frac{F}{A}$$

- P - Pressure (Pa)
- F - Force (N)
- A - Area (m^2)

$$1mbar = 100Pa$$

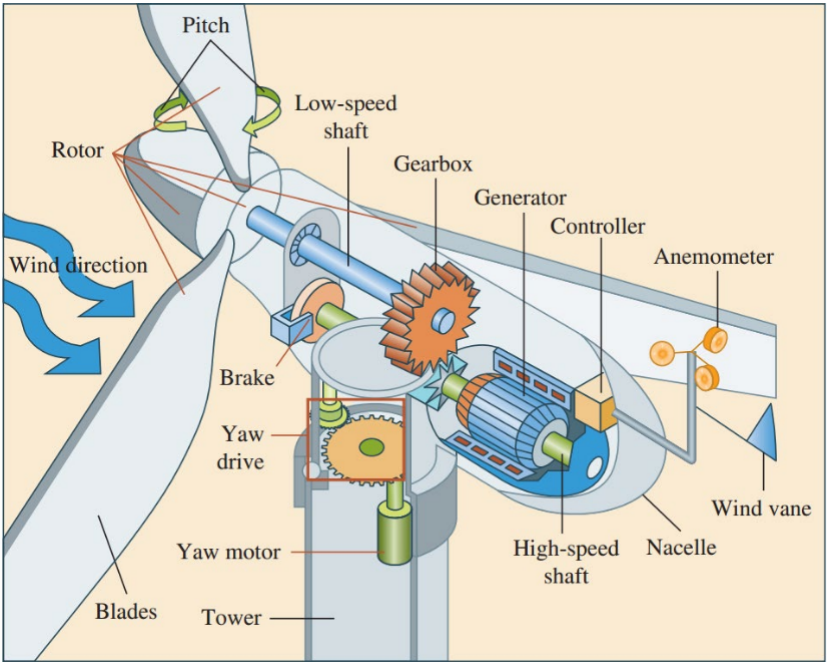
$$1atm = 101320Pa$$

$$1mmHg = 133.3Pa$$

$$1psi = 6895Pa$$

A sea breeze is formed in a temperature difference between the water & the land. The laminar flow is when a liquid (or gas) moves in a straight line. The opposite of this is turbulent flow. The change in velocity behind the turbine creates a turbulent zone, the zone is expandend by this air mixing with the normal air speed.

Wind Turbine:



$$P_w = \frac{1}{2} A \rho_a v_w^3$$

- P_w - Power Generated by the turbine (W)
- A - Area cut by the turbine blades (m^2)
- ρ_a - Density of air ($\frac{kg}{m^3}$)
- v_w - Velocity of the wind ($\frac{m}{s}$)

Betz's Law:

Betz's Law describes the theoretical maximum efficiency of a turbine (59.3%). However, the real efficiency will be much lower in real-world applications, through friction.

Horizontal Axis Wind Turbine:

A HAWT is the most common type of turbine. For it to bo connected straight to the grid the frequency of AC output must match the grid's. In Australia this is 50Hz. To do this different methods are used:

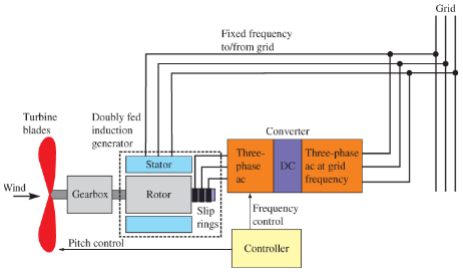
1. Generator Shaft Speed.

$$n_s = \frac{120f}{N_p}$$

   - n_s - RPM
   - f - Frequency (Hz)
   - N_p - No. of Poles

2. Frequency Converter:
   AC->DC->AC->Act Transformer

3. Double-fed



Wind Farms:

The turbines offer the advantage of scale. Containing the ability to share maintainance.
Offshore wind farms are gaining popularity. This is due to no obstacles in the way (e.g. mountains or houses) and uniform windspeeds (e.g. again nothing in the way).

Pitch & Yaw:

Pitch control allows controlling the angle the blade to control the turbine speed. Usually controlled by hydraulic actuaters in a 4-20mA loop. Yaw control points the direction of the turbine. It orients the "centre" of wind by the blades before the nacelle.

Drive-train:

Most turbines rotate @ 20rpm. The gearbox increases this to 1500-1800rpm.

Week 9 - Hydro Power:

Types of Hydro:

Dam

Water from the dam flows through penstocks. They direct it ot the turbine to the generate power and electricity.

River

Uses the K_E of the running water to generate the electricity.

Types of Turbines:

The first type is the impulse turbine. It is primarily used in applications with high pressure & low flow rates. The second type is the reaction turbine. These are usually submerged in water and develop a torque from differential water pressures. The third type is cross flow turbines. These have the water flow across the turbine.

Tidal:

$$W_{PE} = \frac{1}{2}\rho A g h^2$$

- W_{PE} - Water's potential energy (J)
- ρ - Density of water ($\frac{kg}{m^3}$)
  - Saltwater - $1220\frac{kg}{m^3}$

  - Fresh water - $1000\frac{kg}{m^3}$
- A - Area (m^2)
- g - Gravity ($9.8\frac{m}{s^2}$)
- h - $\frac{1}{2}h_{max}$

Conversion Factors:

Some Useful Conversion Factors

To convert from	To	Multiply by
foot (ft)	meter (m)	0.3048
cubic feet (ft^3)	cubic meter (m^3)	0.02832
gallon (US)	cubic meters (m^3)	0.003785
gallon (US)	cubic feet (ft^3)	0.1336
cubic feet per second (ft^3/s)	cubic meters per second (m^3/s)	0.02832
pounds per square foot (lb/ft^2)	newtons per square meter (N/m^2)	47.88
foot-pounds per second (ft-lb/s)	kilowatt (kW)	0.001355

Week 10 - Hydrogen, Fuel Cells and Enenrgy Storage:

Hydrogen:

Hydrogen is the most common element in the world (75% of the mass). Focusing upon earth, hydrogen can be found everywhere and nearly all living things. However, hydrogen is very rarely in pure gaseous form (1ppm). As such hydrogen must be extracted from larger molecules.

The oxidation (burning) of hydrogen generates thermal energy. Unlike other form of fossil fuel combustion (complete CO_2 or incomplete CO), hydrogen only form water. Since burning hydrogen does not produce visible light it is much more efficient. Hydrogen generates 120-142$\frac{MJ}{kg}$. However, to store hydrogen is must be cooled to almost 0K.

Hydrogen Colour Spectrum:

Hydrogen is colour-less gas, however there is a colour-coded table (below) to identify the method of production.

Colour code	Manufacturing process
Green Hydrogen	Electrolysis of water powered by renewable energy
Blue Hydrogen	Steam reforming of natural gas with CCS#
Grey Hydrogen	Steam methane reformation without capturing GHG*
Black and Brown Hydrogen	Use of Black or Brown Coal for H_2 production
Pink Hydrogen	Electrolysis of water powered by nuclear energy
Turquoise Hydrogen	Methane pyrolysis

#CCS – Carbon Capture and Storage; *GHG – Green House Gases

Electrolysis of Water

This utilizes an electrical current through water to create pure hydrogen and oxygen.

Blue Hydrogen

Created through the steam reformation of a natural gas (usually methane) in the presence of the a catalyst.

$$CH_{4(g)} + H_2O_{(g)} <-> CO_{(g)} + 3H_{2(g)}$$

$$CO_{(g)} + H_2O_{(g)} <-> CO_{2(g)} + H_{2(g)}$$

Fuel Cells:

A fuel cell is a device that converts electrochemical energy from a substance into a direct current. This is similar to a battery, bu the fuel can be stored outside of the reaction area. The chemical reaction (oxidation) occurs releasing the electron and producing electricity. Just remember the HFC and LEFC from chemistry research.

REDOX Reactions

A reactant loses electrons (oxidation) and the other gains the electrons (reduction). These electrons travel through a wire generating a current. For the charges in the cell, remember anodes are negative and cathodes are positive.

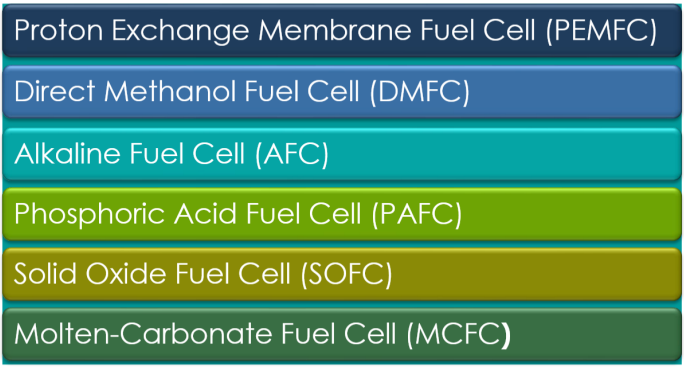
- Oxidation
- Is
- Loss
- Reduction
- Is
- Gain

There are three main fuel cell system:

1. Fuel Cell Stack: A group of fuel cell connected together to increase the outputted power.
2. Fuel Processing Unit: The process the converts the fuel into usable form for the reaction.
3. Heat Exchange Unit: Captures some excess heat generated during the reaction.

Types of the Fuel Cells

The fuel cells describe the chemistry and physical makeup of the cell.



Primary & Secondary Energy:

Primary energy is the naturally found energy souces. These include oil, coal, waste, biomass, or wind. Once these undergo a transformation into a usable form, they become secondary energy sources. Petroleum products, electricity or biofules.

Energy Storage:

The storage of excess energy is an essential link in the energy supply chain. The most basic form is batteries for renewable energy. This is because they supply intermittent outupts, creating an unrealiable form of energy supply. Energy storage stores the excess energy when it is generated and suppliment when extra power is required.

Mechanical - Flywheel

The flywheel is stored in a low friction environment. It smoothes out power and continues generation through the K_E being stored in its rotation. To connect it to the grid, and electrical machine used intially as a motor to convert electricity into movement and then power is needed, it is used as a generator.

Mechanical - Pumped Hydro

Just remeber the hydro from Week 9. Here there are two main types.

1. Open-loop: Lower reservior is part of a running river. The upper reservior is an extra lake.
2. Closed-loop: The upper and lower reservior are two seperate system, not part of running water.

Electrochemical - Batteries

Battery Technology	Energy Density (Wh/kg)	Power Density (W/kg)	Cycle Efficiency (%)	Cyclability (Cycles)	Energy capital cost ($/kWh)	Power capital cost ($/kW)
Lead-Acid (LA)	30 - 50	75 - 300	70 – 80	200 - 2000	200 - 400	300 - 600
Lithium-Ion (Li-Ion)	100 - 200	150 - 315	75 - 90	500 - 2000	600 - 2500	1200 - 4000
Nickel Metal Hydride (NiMH)	60 - 80	80 - 300	66	< 3000	360	180
Nickel Cadmium (NiCd)	50 - 75	150 - 300	60 - 83	500 - 2000	800 - 1500	500 - 1500
Vanadium Redox (VRB)	10 - 30	80 - 150	75 - 85	> 16000	150 - 1000	600 - 1500
Zinc-Bromine (ZnBr)	30 - 50	100	66 - 80	> 2000	150 - 1000	700 - 2500

Week 11 - Grid Connection:

Conventional Electricity Grid:

A conventional grid contains large, centralized power generation. These can include generators (coal, LNG or wind) or solar. Large transmission lines are

connected to substations which in turn connect the local housing infrastructure. These substations are also capable to stepping down the voltage to usable levels. Conventional grid infrastructure are designed primarily for fossil fuels and thus are uni-directional and centralized by design.

Three-phase AC:

Generally electricity is generated in a three-phase waveform. These waveforms are sinusoidal, and with a phase shift of 120°. The reason that thre-phase power is used is that it is economical to produce and transit over single-phase. By using high voltage within a system, current can be reduced saving on the conducter diameter within a cable.

Types of Power:

- True Power (P): This is the active component power that does the work. It is measured in Watts
- Reactive Power (Q): It is the rate at which stored energy in an inductor or capacitor is returned to the source. Measured in Volt-Ampere Reactive (VAR), and does not do any useful work. The only bit it does in flow back and forth between the source and the load.
- Apparent Power (S): The product of a root mean sqaure voltage and current. This is measured in Volt-Amperes (VA).

Power Factor

The PF is the cosine of the angle between the voltage and current phasors. The PF is a useful component in calculationg the True Power within a system.

$$PF = \frac{TruePower(P)}{ApparentPower(S)}$$

When providing an answer to the equation it is usually accompanied by the word lagging or leading, depending on whether the lags the voltage or leads the voltage.

Power in Three-phase Circuit:

$$P_T = \sqrt{3} V_L I_L \cos(\phi)$$

Where:

- P_T - True(active) power (in W)
- V_L - Line voltage (V)
- $I - L$ - Line current (A)
- cos(ϕ) - The power factor

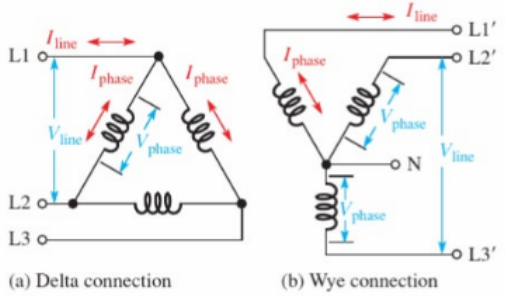
Three-Phase Connections:

At the source side, the three-phase transformer winding are connected in a Y or Δ. The transformers will contain a primary and secondary windings. This can be mimicked on the load side.
In a Δ connection:

$$I_{Phase} = \frac{I_{line}}{\sqrt{3}} \text{ and } V_{Phase} = V_{line}$$

In a Y connection:

$$I_{Phase} = I_{line} \text{ and } V_{Phase} = \frac{V_{line}}{\sqrt{3}}$$



Power Transmission:

Power transmission involves moving energy around, such as from a source to a load. To do this it requires the installations of substations between the source and load. Based upon where the substations are placed, they will be called transmission or distribution substations.

Substations:

A substation typically consists of a network of switching equipment, control equipment and transformers to change the voltage. Closer to the source a transmission substation will be utilized to step up the voltage for easier/better transportation. Whereas a distribution substation is used to connect a load, and consists where the voltage will be stepped down for distribution.

Week 12 - The Effects of Renewable Energy:

Intergrating Renewable Energy Sources to the Grid:

Positives

- No greenhouse gas emissions or air pollutants
- For small scale operations, it has a lower initial costs
- Allows new creation of jobs
- As long as the base is there, the renewable can be used, e.g. with solar, as long as there is generally sunshine
- The triple bottom line is generally considered to be good

Negatives

- Power quality of the source, such as 230V @ 50Hz when generating the power
- Variation and intermittency of the power generation, e.g. solar having maximum power during midday and zero power during the night
- The best source of power generation is always different based upon where the location of them is being placed.
- Forecasting the future amount of power based upon the weather data/trends
- Making a large system to supplement/replace and entire grid will becomes very expensive, very quickly

Grid Connection Criteria:

The voltage and frequency of the soucre must match with the grid's voltage, frequency and phase. This all combines into the power quality, where the waveform must also match or be changed to matched the grids. Reducing the amount of the harmonics present within the power source.

Harmonics

Harmonics are produced at the source where electricity is produced or used.

Noise

RF noise may be induced in power lines due to high frequency switching of equipment in the vicinity. To remove harmonic components and noise, a filter can be used to extract specific aspects from the voltage waveform.

Low Voltages Ride Through:

This is considered when the voltage is <90

Flicker:

Flicker is a momentary drop in system voltage level. This can only be for a few milliseconds. Flicker is most notable through standalone renewable energy systems, but an inverter can be used to control and limit the amonut of flicker within an electricity network.